

Uncovering Critical Thinking Disposition Profiles in Adolescents Through Fuzzy Clustering: Developmental Implications for Video Literacy and Learning Analytics

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Abstract

This article investigates how integrated configurations of critical thinking dispositions, combining self efficacy, epistemic beliefs, and reasoning style, influence educational video evaluation and school related usage decisions among 1,079 secondary school students. Fuzzy C-Means clustering identified three distinct dispositional profiles at each educational level. In high school, two high disposition subtypes emerged: an intuitive profile (high self efficacy, low analytical reasoning) and an analytical profile (moderate self efficacy, high analytical reasoning). Only the analytical profile linked to superior critical performance and confirmation bias resistance. The intuitive profile, despite high confidence, matched low disposition students. In middle school, performance scaled with overall disposition intensity, without reasoning style moderation. Findings indicate that effective high school critical thinking requires motivation analytical reasoning coupling, a late adolescent threshold. The study informs differentiated pedagogy and computer-based decision support systems (e.g., learning analytics platforms) in digital education.

Keywords: Critical thinking dispositions, Video literacy, Self-efficacy, Epistemic beliefs, Cognitive reflection, Analytical reasoning, Fuzzy C-Means clustering, Person-centred approach, Learning Analytics, Adolescent development, Digital literacy, Educational decision-making

1 Introduction

Video consumption, particularly on platforms such as YouTube, has grown exponentially among adolescents (Twenge, Martin, & Spitzberg, 2019), reflecting profound changes in their media practices. Students increasingly prefer videos to texts for autonomous learning (Livingstone & Blum-Ross, 2020; Vogels, Gelles-Watnick, & Massarat, 2022). This shift to audiovisual environments creates educational challenges, especially the spread of misleading or erroneous information (Kiili, Coiro, & Rääkkönen, 2019). In this context, identifying reliable content forms a core, complex component of digital literacy (Potocki et al., 2020; Rutter, Blinzler, Ye, Wilson, & Twidale, 2019).

Critical thinking, defined as the ensemble of dispositions and skills that enable the evaluation of the epistemic quality of information to determine the degree of trust warranted (Chen, Wang, & Zheng, 2024; Ennis, 1996; P. Facione, 1990), plays a foundational role in knowledge construction (Binkley et al., 2011). Although research emphasises the importance of developing these competencies from an early age (Chen et al., 2024), their effective implementation depends less on procedural mastery than on the voluntary mobilisation of stable cognitive and motivational dispositions (Ennis, 1996; P. Facione, 1990; P.A. Facione, Sanchez, Facione, & Gainen, 1995; Halpern, 2013). This distinction between competence and disposition underpins contemporary approaches to critical thinking: possessing evaluative skills does not guarantee their activation when students encounter online information.

In France, this awareness has led to the integration of the critical evaluation of digital content into school curricula. Since 2016, the common framework of knowledge, skills, and culture (*Socle commun de connaissances, de compétences et de culture*¹) has provided middle school students with a shared set of knowledge, skills, values, and attitudes expected to be mastered by the age of 16. Since 2019, the PIX certification² further extended this agenda to high school, promoting and certifying students' digital competences, including critical information literacy skills such as evaluating the credibility of online content and its sources. This institutional differentiation between middle and high school, both in terms of curricular expectations and certification frameworks, provides a particularly relevant context for examining how critical thinking dispositions develop across adolescence, and constitutes one of the motivations for the comparative design adopted in the present study.

Yet, videos present specific cognitive challenges that set them apart from textual documents: their transient and multimodal nature places considerable demands on working memory, often leading to superficial evaluation strategies that neglect argument quality or speaker expertise (Mayer, 2002; Merkt, Ballmann, Felfeli, & Schwan, 2018). These features make the critical evaluation of videos a qualitatively distinct epistemic task, yet research has primarily focussed on textual documents (Abed & Barzilai, 2023), leaving the links between critical thinking dispositions and video processing poorly understood (Carmichael, Reid, & Karpicke, 2018). This gap limits, for instance, the design of individualised pedagogical support tailored to the specific demands of audiovisual information environments (Yoon, Lee, & Jo, 2021); this aligns

¹<https://www.education.gouv.fr/le-socle-commun-de-connaissances-de-competences-et-de-culture-12512>.

²<https://pix.org/en/certify-my-digital-skills>.

with computer-mediated learning environments where analytics enable profile-based adaptation.

Addressing this gap requires examining not only which dispositions matter, but also how they combine into integrated profiles that differentially shape students' video evaluation processes. The present study is part of the POUCEEC (*Parcours Ouverts et Compétences pour l'Éducation à l'Esprit Critique*) project³, funded by the French Ministry of National Education and aimed at evaluating the effectiveness of a digital platform designed to foster critical thinking through video analysis. The study seeks to deepen the understanding of the cognitive processes and individual factors that influence how students evaluate online videos. This work extends the findings of Hémon, de Checchi, Arguel, Trémolière, and Amadiou (2025), who demonstrated that certain dispositions, namely self-efficacy and epistemic beliefs, predict students' ability to discriminate between videos based on source expertise.

Compared to the reference paper (Hémon et al., 2025), the present study differs in four essential respects:

1. *Methodology*: we adopt a person-centred, rather than variable-centred approach, which aims to characterise integrated configurations of cognitive and motivational dispositions and to assess their impact on both epistemic judgements and educational usage decisions regarding online videos. To this end, we employ the Fuzzy C-Means (FCM) clustering method (Bezdek, 1981), which remains underutilised in educational research and is particularly suited to capturing the gradual, overlapping nature of critical thinking dispositions.
2. *Theoretical design*: we extend the scope of evaluation beyond credibility judgements to include the quality of students' educational usage decisions, specifically their inclination to select expert-authored over non-expert videos for school presentations. We also introduce reasoning style as a theoretically motivated moderator of the relationship between critical thinking dispositions and evaluative performance, a dimension absent from the reference study. This dual extension is particularly relevant in learning contexts, where the ability to make sound epistemic choices has direct consequences for knowledge construction.
3. *Developmental perspective*: we adopt an explicit developmental perspective by comparing middle and high school students, moving beyond a global comparison to examine how critical thinking disposition profiles evolve with cognitive maturation. Specifically, we hypothesise that the moderating role of reasoning style becomes increasingly central in late adolescence, as the coupling of motivation and analytical thinking gradually emerges as a prerequisite for effective critical performance.
4. *Sample size*: Although based on the same experimental protocol, our study relies on a distinct and substantially larger sample ($N = 1,119$ vs. 363). This increase in sample size is not merely quantitative: it is a prerequisite for the person-centred, profile-based analyses employed here, which require sufficient within-group variance to yield stable and interpretable cluster solutions.

³<https://pouceec.fr/>.

Together, these contributions position the present study at the intersection of epistemic cognition, developmental psychology, and computers in education. They are operationalised through three research questions:

- RQ1: Do critical thinking dispositions differ significantly between middle and high school students?
- RQ2: What profiles of critical thinking dispositions can be identified in the analysed population?
- RQ3: To what extent do these profiles influence (a) judgements of information credibility and source trustworthiness, and (b) the quality of educational usage decisions regarding videos among middle and high school students?

These RQs address the limitations of variable centred approaches by prioritising profile interactions and developmental thresholds.

The remainder of this article is organised as follows. Section 2 presents the theoretical framework and prior research on the critical evaluation of digital content and the dimensions of critical thinking dispositions. Section 3 describes the methodology, including the educational context, participants, data collection procedures, and analytical methods. Section 4 reports results on developmental differences, critical thinking disposition profiles, and their role in shaping students' epistemic judgements and educational usage decisions. The discussion in Section 5 interprets these findings in light of developmental and epistemic cognition theories. Section 6 outlines pedagogical and methodological implications for curriculum design, teacher education, and digital learning tools. Section 7 discusses the limitations of the study and proposes directions for future research.

2 Background

2.1 Digital content and videos: between bounded rationality and epistemic demands

Given the vast amount of information available online, individuals often have to assess the credibility of the content they encounter (Metzger & Flanagin, 2013). This cognitive demand operates within a framework of bounded rationality (Simon, 1956), where limited attentional and working memory resources constrain the depth of information processing. Under these conditions, individuals frequently employ heuristic strategies, so-called satisficing approaches, that prioritise efficiency and cognitive economy over exhaustive analysis (Fu & Pirolli, 2007; Pirolli & Fu, 2003). While adaptive in many contexts, such strategies increase vulnerability to systematic biases, especially when the information is complex, conflicting, or deliberately misleading (Tversky & Kahneman, 1974). These vulnerabilities are particularly pronounced in the context of online videos, which impose additional cognitive constraints that further reduce the likelihood of deep epistemic processing, as detailed below.

More specifically, critical evaluation of digital content relies on two interdependent types of judgements (Abed & Barzilai, 2023; Hémon et al., 2025; Kiili et al., 2019; Kohnen, Dawson, & Mertens, 2022). The first concerns judgements of information

credibility, that is, the perceived quality of the message in terms of accuracy, completeness, and evidentiary justification. The second concerns judgements of the source's epistemic trustworthiness, focussed on perceived expertise, integrity, and benevolence. These two dimensions are interdependent in the sense that assessments of source credibility typically inform and constrain judgements of message quality, and vice versa (Hémon et al., 2025). Together, they underpin what Sperber et al. (2010) describe as contextualised epistemic vigilance, namely the capacity to regulate one's degree of trust in information as a function of both its content and its source. This capacity is central to the present study, as we examine how students' critical thinking disposition profiles shape precisely these two types of judgement, as well as the educational usage decisions that follow from them.

Online videos represent the dominant information format among adolescents (Livingstone & Blum-Ross, 2020; Ofcom, 2023), yet they impose specific cognitive constraints that go beyond those associated with textual documents. Their transient nature limits the possibility of re-reading and reflection, placing considerable demands on working memory (Mayer, 2002), while their multimodal character, combining voice tone, visual appearance, and narrative style, fosters the emergence of affective heuristics (Aljalabneh, 2024; Sundar, Molina, & Cho, 2021). These heuristics often lead individuals to prioritise surface-level cues, such as fluency of speech or perceived likability, over a rigorous assessment of epistemic validity (Irwan, 2024; McGrew, Breakstone, Ortega, Smith, & Wineburg, 2018). In this context, affect itself becomes a source of information, actively shaping judgements in ways that may bypass analytical processing altogether (Jonas, Kerwer, Chasiotis, & Rosman, 2024; Schwarz, Jalbert, Noah, & Zhang, 2021). Against this backdrop, critical thinking dispositions (CTDs) play a crucial moderating role: they determine not only the ability to identify relevant epistemic cues but, more importantly, the willingness to invest the cognitive resources necessary for their thoughtful processing. As Hémon et al. (2025) emphasise, these dispositions vary across three interrelated dimensions, namely motivational (self-efficacy), epistemic (schemes of knowledge justification), and cognitive (reasoning style). Understanding how these dimensions combine into integrated profiles, rather than operating in isolation, is therefore a theoretical challenge addressed in the following section.

2.2 Profiles of critical thinking dispositions in video evaluation

Critical thinking does not reduce to a set of technical skills; it requires the integrated mobilisation of motivational, epistemic, and cognitive factors (Ennis, 1996; P. Facione, 1990). In the context of online video evaluation, as Hémon et al. (2025) synthesise, drawing on a broader body of research, these dispositions vary across three interrelated dimensions.

1. **Self-efficacy (SE):** Defined as the belief in one's ability to successfully perform a specific task (Bandura, 1997), SE acts as a foundational motivational lever. High SE promotes engagement in cognitively demanding tasks (Hayat, Shateri, Amini, & Shokrpour, 2020; Tang, Tseng, & Tang, 2022) and predicts more accurate video evaluations (Hémon et al., 2025). However, this relationship is not linear: when self-efficacy is poorly calibrated, that is, when confidence in one's evaluative abilities

exceeds actual competence, it can lead to epistemic overconfidence. This manifests as a tendency to underestimate task complexity and to forgo verification strategies, ultimately impairing rather than supporting critical performance (Anttonen, Räikkönen, Kiili, & Kiili, 2024; Dunlosky & Rawson, 2012; Pennycook et al., 2021).

2. **Epistemic beliefs (schemes of knowledge justification):** Evaluating a video also involves activating beliefs about what constitutes valid knowledge. Three justification schemes consistently emerge in the literature (Anmarkrud, Andresen, & Bräaten, 2019; Hämäläinen, Kiili, Räikkönen, & Marttunen, 2021):

- *Justification by authority (JA)*: reliance on the expertise of the author, which is also referred to as academic authority (Bromme, Kienhues, & Porsch, 2010);
- *Justification by multiple sources (JMS)*: seeking convergence across independent sources as a means of establishing the validity of a claim;
- *Justification by personal knowledge (JPK)*: anchoring validity in personal experience or prior beliefs.

These justification strategies are not mutually exclusive, but they entail distinct epistemic practices: a strong preference for personal justification is associated with limited sourcing, whereas valuing source multiplicity promotes corroboration behaviours (Barzilai, Tzadok, & Eshet-Alkalai, 2015).

3. **Reasoning style:** this dimension is measured through the Cognitive Reflection Test (CRT) (Frederick, 2005) and its developmental version (CRT-D) (Young & Shtulman, 2020), which measures the tendency to inhibit intuitive responses in favour of more deliberate reasoning. In the video context, this distinction predicts the ability to resist surface-level biases (Gervais, 2015; Pennycook & Rand, 2019). Information processing oscillates between two modes: a fast, intuitive, and heuristic *System 1* with low CRT, and a slow, analytical, and reflective *System 2* with high CRT (Evans & Stanovich, 2013; Kahneman, 2011). We therefore hypothesise that reasoning style does not operate in isolation but acts as a moderator of the relationship between self-efficacy, epistemic beliefs, and the quality of video evaluation. Crucially, we expect this moderating role to become increasingly pronounced in late adolescence, as the cognitive capacity to inhibit heuristic responses gradually matures. This developmental hypothesis is central to the profile-based analyses conducted in the present study, where we examine whether the coupling of high self-efficacy with analytical reasoning constitutes a distinct and functionally superior dispositional configuration, and whether this coupling emerges specifically in late adolescence by contrast with middle school students.

Taken together, these three dimensions, namely self-efficacy, epistemic beliefs, and reasoning style, do not operate independently but combine into integrated dispositional configurations that shape how students approach the evaluation of online videos. Identifying such profiles is essential to distinguish students whose confidence is grounded in analytical reasoning from those whose confidence rests on uncritical intuition, and to examine how the interaction between self-efficacy, epistemic beliefs, and reasoning style translates into different patterns of judgement and decision-making. This perspective is particularly relevant for informing pedagogical design and personalising

digital learning tools, as educators and adaptive systems interact not with isolated variables but with individuals who embody integrated dispositional configurations.

2.3 Toward an original contribution: profiling, decision-making, and developmental differentiation

Hémon et al. (2025) represent a significant advance in the understanding of how critical thinking dispositions shape students' evaluation of online videos, by demonstrating that certain CTDs predict students' ability to discriminate between videos based on source expertise. Nevertheless, three structural limitations warrant the present extension.

1. Methodological: their variable-centred approach treats each CTD dimension in isolation, and therefore fails to capture the complex interactions among them. This precludes the identification of integrated dispositional profiles, which are essential for understanding how motivation, epistemic beliefs, and reasoning style jointly shape students' capacity to critically assess online videos;
2. Operational: their analyses focus on evaluation scores alone, without examining the quality of educational usage decisions, that is, whether students preferentially select expert-authored over non-expert videos for school-related purposes. This dimension is crucial in learning contexts, where the ability to identify reliable sources has direct consequences for knowledge construction;
3. Developmental: their comparison between middle and high school students remains at an aggregate level, reporting mean differences without exploring how CTD profiles are structured differently across educational levels, nor whether the moderating role of reasoning style varies with cognitive maturation.

The present study addresses these limitations in three ways. First, it identifies CTD profiles through Fuzzy C-Means clustering, thereby adopting a person-centred perspective. Second, it assesses their impact not only on evaluative judgements but also on educational usage decisions. Third, it analyses how dispositional profiles and their functional consequences differ between middle and high school students, with particular attention to the moderating role of reasoning style. The following section describes the methodological framework through which these objectives are operationalised.

3 Methodology

3.1 Ethical Statement

The study was conducted as part of the POUCEEC project, introduced in Section 1, funded by the French Ministry of National Education. It strictly adheres to ethical guidelines governing research involving minors in educational settings. As part of the POUCEEC project, the research protocol was approved by the Research Ethics Committee of the University of Toulouse (France) and was formally validated in advance by the relevant academic authorities.

All participants received age-appropriate information about the study's objectives, duration, and procedures. Informed consent was obtained from both students and their

252 legal guardians prior to data collection. Participants were explicitly assured of their
 253 right to withdraw at any time without any negative consequences for their academic
 254 progression. All data were collected anonymously; no personally identifying information
 255 was recorded. Only random identifiers and aggregated sociodemographic variables
 256 (grade level, gender, age) were retained. Data storage complied with the General Data
 257 Protection Regulation (GDPR) and was carried out on a secure university server.

258 3.2 Educational context

259 The study involved a network of 17 public secondary schools (11 middle schools, 6
 260 high schools), recruited in collaboration with regional academic authorities. The final
 261 sample comprised 1,119 students ranging from age 11 to age 18 (French *6ème* to
 262 *Terminale*). Participants were excluded on two grounds: those with excessive missing
 263 data or aberrant response patterns (e.g., invariant responses or clear contradictions
 264 with task instructions) ($n = 23$), and outliers identified through a dedicated detection
 265 procedure described in Section 3.5.1 ($n = 17$), yielding a final analytic sample of 1,079
 266 students. This sample was distributed as follows: 776 middle school students (mean
 267 age = 12.88; SD = 1.27) and 343 high school students (mean age = 16.21; SD = 1.16).
 268 Table 1 provides an overview of participants by school level and gender.

Table 1 Participants by school level and gender

Genre	Middle school	High school
Female	413	172
Male	346	157
Other	9	6
PNS	8	8

PNS = prefer not to say

269 This cohort differs from that utilised by Hémon et al. (2025) ($n = 363$), despite
 270 both originating from the same research programme. Our larger sample enables more
 271 robust analyses (Gibson, Hu, Huisman, & Barratt, 2017; Rachman P, Mangkuto A,
 272 Koerniawan Donny, Sarwono, et al., 2024; Serdar, Cihan, Yücel, & Serdar, 2021), par-
 273 ticularly within a person-centred framework (Sect. 3.5.3), with cluster sizes (81 - 325)
 274 exceeding minimums for fuzzy clustering stability (Dalmaijer, 2023; Dalmaijer, Nord,
 275 & Astle, 2022) and interpretability ($>10\%$ N) (Aschenbruck & Szepannek, 2020a), as
 276 confirmed by high between-cluster discriminability ($\varepsilon^2=0.30 - 0.73$).

277 3.3 Study design and data collection

278 The study employed a *within-subject* experimental design, closely replicating the pro-
 279 tocol of Hémon et al. (2025) but enriched by an explicit focus on educational usage
 280 decisions. Data obtained via a questionnaire were gathered in person, with teach-
 281 ers present, during regular school hours in a single session lasting 50 to 60 minutes,

282 using the computer-based platform (LimeSurvey). The questionnaire unfolded in three
 283 sequential parts:

- 284 • Part 1 (**P1**): *Sociodemographic questionnaire*. Students reported their age, gender,
 285 grade level as illustrated in Table 1. Gender was not employed as a discriminant
 286 indicator in this study; however, it will be implemented in subsequent investigations
 287 as illustrated in the conclusion (Sect. 7).
- 288 • Part 2 (**P2**): *Evaluation of experimental videos*. Students watched four videos (45–
 289 55 seconds each) addressing the socio-scientific issue regarding the reduction of meat
 290 consumption to facilitate the growth of organic farming (Mazac et al., 2025). The
 291 four videos were created for an earlier research project that examined students’
 292 assessments of conflicting videos (Lescarret et al., 2024). They consisted of purpose-
 293 built interviews in which the interviewee’s positions on the issue and their levels of
 294 expertise were systematically varied. The experimental design orthogonally crossed
 295 two factors: source expertise (expert versus non-expert) and stance on the issue
 296 (in favour of versus against reducing meat consumption). A banner was displayed
 297 at the bottom of the screen at the start of each video, showing the interviewee’s
 298 name and age in the non-expert condition, and the name, occupation (e.g. agronomy
 299 researcher), and professional affiliation in the expert condition. In order to under-
 300 score the disparity in proficiency, experts were interviewed in an office setting, while
 301 non-experts were interviewed while shopping at a supermarket.
 302 The four resulting video conditions are detailed in Table 2. For example, V_1 features
 303 a non-expert in favour of meat reduction, while V_4 features an expert against it.
 304 To mitigate order effects, the video sequence was counterbalanced across classes;
 305 specifically, the videos were distributed randomly.

Table 2 Evaluation of experimental videos

Attitude	Non Expert	Expert
In favour	Video 1 (V_1)	Video 2 (V_2)
Against	Video 3 (V_3)	Video 4 (V_4)

306 After each video, students rated on 9-point Likert scales (1=not at all/very low;
 307 9=completely/very high): (i) the epistemic trustworthiness of the source (perceived
 308 expertise, integrity, benevolence) (6 questions); (ii) the information credibility of
 309 the source (accuracy, completeness, justification) (6 questions); (iii) their intention
 310 to use the video for a school presentation (1 question).

- 311 • Part 3 (**P3**): *Measurement of critical thinking dispositions* (Sect. 2.2). In this part,
 312 students answered several questions assessing: (i) self-efficacy (SE) specific to video
 313 evaluation (4 questions); (ii) epistemic beliefs (12 questions) structured into three
 314 subscales: justification by authority (JA), multiple sources (JMS), and personal
 315 knowledge (JPK); (iii) reasoning style, measured via a composite version of the
 316 Cognitive Reflection Test (CRT) (Frederick, 2005) (5 questions) and its develop-
 317 mental adaptation (CRT-D) (Young & Shtulman, 2020), tailored for adolescents (3
 318 questions).

319 For further details on the questionnaire, the reader is referred to Hémon et al.
320 (2025).

321 3.4 Metrics and operationalisation of variables

322 The operationalisation of variables and their corresponding metrics was organised
323 into two main categories: *i*) video-related judgements (P2) and *ii*) critical thinking
324 dispositions (P3):

325 3.4.1 Video-related judgements

326 For each of the four videos (V_1, V_2, V_3, V_4), we computed the following indicators:

- 327 • Epistemic trustworthiness of the source (**ET**): a composite metrics computed as the
328 average rating of the 6 questions evaluating perceived source expertise, integrity,
329 and benevolence;
- 330 • ET_{score} is an accuracy score derived from ET. Because ET is measured on a 9-point
331 scale, the midpoint (5) corresponds to a neutral evaluation: values above 5 indicate
332 an overall positive trustworthiness judgement, whereas values below 5 indicate a
333 negative trustworthiness judgement. Students received 1 point when they assigned
334 a positive trustworthiness judgement ($ET > 5$) to an expert-authored video (Videos
335 2 and 4), and 1 point when they assigned a negative trustworthiness judgement (ET
336 < 5) to a non-expert video (Videos 1 and 3). The ET_{score} is the sum of these points
337 and therefore ranges from 0 (judgements all incorrect) to 4 (judgements all correct).
- 338 • Information credibility (**IC**): the average rating of the 6 questions evaluating the
339 accuracy, completeness, and justification of the message;
- 340 • School usage intention (**SU**): a single-question score reflecting the student's stated
341 likelihood of using the video for a school presentation.
- 342 • School usage intention quality (SU_{score}): a global performance score indicating the
343 extent to which students preferentially selected expert-authored videos over non-
344 expert videos for school usage. This index was computed by assigning a binary value
345 to each video choice (1 = selected an expert video; 0 = selected a non-expert video)
346 and summing these values across the four videos.

347 It should be noted that IC_{score} could not be computed in this study, as we
348 prioritised analysing the direct effects of perceived expertise and did not include an
349 objective benchmark of information quality, a criterion that students could not reliably
350 evaluate directly.

351 3.4.2 Critical thinking dispositions

352 The following metrics were derived to capture students' critical thinking dispositions
353 (CTD):

- 354 1. Self-efficacy (**SE**): the mean of 4 questions assessing self-efficacy specific to video
355 evaluation.
- 356 2. Epistemic beliefs: the mean of 12 questions assessing epistemic beliefs specific to
357 video evaluation, measured through three subscales (**JA**, **JMS**, **JPk**).

358 3. Reasoning style (*CRT_score*): the total number of correct responses, combining
359 scores from the Cognitive Reflection Test (CRT) and its developmental adaptation
360 (CRT-D).

361 Table 3 summarises all constructed metrics and their abbreviations.

Table 3 Metrics used in the study.

Part	Metrics	Description	Type	Points
Video-related judgements	ET (V_1, V_2, V_3, V_4)	Epistemic trustworthiness of the source (per video)	Mean	[1-9]
	ET _{score}	Epistemic trustworthiness scoring quality for all videos	Sum	[0-4]
	IC (V_1, V_2, V_3, V_4)	Information credibility (per video)	Mean	[1-9]
	SU (V_1, V_2, V_3, V_4)	Intention to use the video for school presentation (per video)	Single value	[0-1]
	SU _{score}	School usage intention quality for all videos	Sum	[0-4]
Critical Thinking Dispositions	SE	Self-efficacy	Mean	[1-9]
	JA	Epistemic beliefs: Justification by authority	Mean	[1-9]
	JMS	Epistemic beliefs: Justification by multiple sources	Mean	[1-9]
	JPK	Epistemic beliefs: Justification by personal knowledge	Mean	[1-9]
	CRT _{score}	Reasoning style (CRT and CRT-D)	Sum	[0-8]

362 According to methodological recommendations in the literature (Little, 2024;
363 Moeller, 2015), all metrics points were standardised using the Proportion of Maximum
364 Scaling (POMS) method, which rescales scores onto a [0, 1] interval. This transforma-
365 tion preserves the rank order of responses while ensuring metric comparability across
366 questions and subscales, thereby enhancing compatibility with standard statistical
367 procedures.

368 3.5 Data Analysis

369 All analyses were conducted using R (version 4.4.2), combining non-parametric com-
370 parative approaches (Kruskal–Wallis test, Mann–Whitney U tests, and Wilcoxon
371 signed-rank tests) with exploratory Fuzzy C-Means clustering (via the `cmeans` func-
372 tion from the `e1071` package, version 1.7.16⁴). Non-parametric methods were preferred
373 due to unequal group sizes (Table 1) and deviations from normality in the distribu-
374 tion of study metrics (Table 3). To control the Type I error rate (i.e., false positives)
375 associated with multiple comparisons (Noble, 2009), p-values were adjusted using the
376 Benjamini–Hochberg procedure (Benjamini & Hochberg, 1995). This correction was
377 chosen because it offers greater statistical power and is less conservative than tradi-
378 tional alternatives, such as Bonferroni or Holm corrections (Glickman, Rao, & Schultz,
379 2014; Ryan, 2021).

380 3.5.1 Data preprocessing and outlier detection

381 An initial exploratory data analysis of the entire dataset ($n = 1,119$) was conducted.
382 Data were first inspected for missing values, logical inconsistencies, and aberrant

⁴<https://www.rdocumentation.org/packages/e1071/versions/1.7-17/topics/cmeans>

383 response patterns. Participants with more than 20% missing data on key variables or
 384 clearly non-serious response patterns (e.g., identical responses across all items, explicit
 385 disagreement with instructions) were excluded ($n = 23$: 20 middle schoolers, 3 high
 386 schoolers).

387 Then, following established practices (Liu, Li, Wu, & Fu, 2019), we conducted
 388 multivariate outlier detection to minimise the disproportionate influence of atypical
 389 response profiles on cluster partitioning and enhance data structure (Nowak-
 390 Brzezińska & Gaibei, 2022). This procedure combined two complementary approaches:
 391 (i) robust Mahalanobis distance, which identifies global deviations from the multi-
 392 variate centroid, and (ii) the Local Outlier Factor (LOF), which compares the local
 393 density of a point to that of its neighbours to detect local anomalies (Liu et al., 2019).
 394 Only participants flagged as outliers by both methods were excluded from the cluster-
 395 ing analysis ($n = 17$: 9 middle schoolers, 8 high schoolers). This conservative criterion
 396 was adopted to minimise the risk of erroneously excluding legitimate response profiles,
 397 ensuring that only participants identified as atypical by two independent detection
 398 approaches were removed (Nowak-Brzezińska & Gaibei, 2022). Results reported in
 399 Section 4 are based on the preprocessed dataset ($n = 1,079$).

400 3.5.2 Non-parametric analyses (RQ1, RQ3)

401 To address RQ1 (*Do critical thinking dispositions differ significantly between middle*
 402 *and high school students?*), we compared the distributions of critical thinking dis-
 403 positions (CTD variables, Sect. 3.4.2) across school levels using Mann–Whitney U
 404 tests.

405 To address RQ3 (*To what extent do these profiles influence (a) judgements of infor-*
 406 *mation credibility and source trustworthiness, and (b) the quality of educational usage*
 407 *decisions regarding videos among middle and high school students?*), we examined
 408 how the identified profiles, defined in Section 4.2, were associated with outcomes from
 409 video-related judgements (ET, ET_{score}, IC, SU, SU_{score}; Sect. 3.4.1). Between-profile
 410 comparisons were conducted using Kruskal–Wallis tests, followed by pairwise Mann–
 411 Whitney U tests with Benjamini–Hochberg correction to control the false discovery
 412 rate (Benjamini & Hochberg, 1995) whenever post-hoc contrasts were warranted. This
 413 non-parametric approach was selected to ensure robust inference while respecting the
 414 structure of our dataset.

415 3.5.3 Fuzzy C-Means clustering analysis (RQ2, RQ3)

416 To address RQ2 (*What profiles of critical thinking dispositions can be identified in the*
 417 *analysed population?*), we performed Fuzzy C-Means clustering separately for mid-
 418 dle and high school students using the CTD variables: SE, JA, JMS, JPK, CRT_{score}
 419 (Table 3). This choice was motivated by the particular structure of our data, as assessed
 420 through the Hopkins statistic. This measure of cluster tendency (Hopkins & Skellam,
 421 1954), computed with the `performance` R package (v0.12.4⁵), was low ($H = 0.35$),
 422 thus indicating a weak natural tendency toward partitioning and fuzzy boundaries

⁵<https://www.rdocumentation.org/packages/performance/versions/0.12.4>

between potential groups. In such cases, hard-clustering methods (e.g., K-means, hierarchical clustering) yield suboptimal performance (Rai, Singh, et al., 2010), whereas fuzzy clustering accommodates the gradual, non-binary nature of critical thinking dispositions. The optimal number of clusters was determined by triangulating multiple validity indices: the Fuzzy Partition Coefficient (Bezdek, 1974) and the average fuzzy silhouette width (Campello & Hruschka, 2006). Solutions with $k = 2$ and $k = 3$ clusters were evaluated. We selected the three-cluster solution as it balanced statistical validity (peak silhouette score) with pedagogical interpretability, a key functional criterion in educational profiling (Aschenbruck & Szepannek, 2020b). The fuzziness parameter (m) was optimised by varying it between 1.25 and 2.5 (Amane, Aissaoui, & Berrada, 2023); $m = 1.75$ yielded the highest fuzzy silhouette index and was retained. Each student was assigned a membership vector (summing to 1) reflecting their degree of belonging to each cluster. The algorithm was run with a maximum of 100 iterations to ensure convergence while maintaining computational efficiency, and random initialisation was repeated 20 times to guarantee solution stability.

For subsequent comparative analyses (RQ3), each student was assigned to the cluster with the highest membership score. This fuzzy clustering approach respects the continuous and overlapping nature of CTDs, avoids overly rigid categorisation, and provides a meaningful, interpretable foundation for analysing how CTD profiles relate to students' evaluative judgements and educational usage decisions.

4 Results

This section presents the findings organised by RQ. Descriptive statistics for all metrics and detailed results of statistical tests (RQ1–RQ3) are provided in the supplementary materials (A) to ensure transparency and analytical reproducibility.

4.1 RQ1 - Developmental differences in critical thinking dispositions

As illustrated in Figure 1 and in the Kruskal–Wallis tests reported in Table 4, we observe significant developmental differences between middle and high school students across all CTD metrics, though effect sizes vary considerably.

Table 4 Kruskal–Wallis tests. Comparison between middle and high school. ε^2 denotes effect sizes.

Metric	χ^2	df	p	ε^2
SE	7.37	1	< .01	.01
JA	14.63	1	< .001	.01
JPK	13.44	1	< .001	.01
JMS	3.86	1	= .049	.003
CRT _{score}	217.91	1	< .001	.19

For all metrics, high school students report higher scores than middle schoolers (both in means and medians). The largest effect is observed for cognitive reflection (CRT_{score}), suggesting substantial cognitive maturation between early and late adolescence. In contrast, although statistically significant, the effect for JMS is negligible ($\varepsilon^2 = 0.003$), highlighting persistent difficulties in systematically deploying corroboration strategies, even among high school students, a finding consistent with Anmarkrud et al. (2019).

These developmental gaps reinforce the hypothesis of a progressive maturation of CTDs, shaped both by cognitive development (Steinberg, 2005) and by repeated exposure to institutional expectations regarding source evaluation (French national curricula, 2016; PIX certification, 2019; Sect. 1).

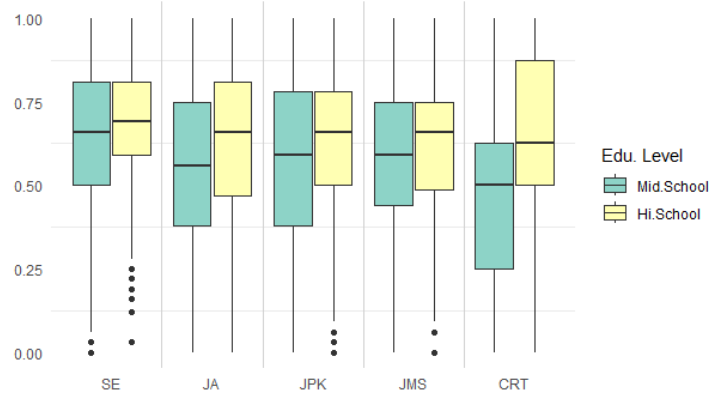


Fig. 1 Distribution of the five metrics selected to identify CTDs (middle school and high school students)

4.2 RQ2 - Identification of critical thinking disposition profiles

Fuzzy C-means clustering revealed three distinct profiles within each educational level (Figure 2, Table 5). These profiles are interpreted in light of the Kruskal–Wallis test results (Table 6).

4.2.1 Critical thinking disposition profiles among middle school students

We identified three distinct profiles:

- Cluster 1 (High CTD, $n = 281$): characterised by high levels of SE (7.33) and elevated scores across all epistemic justification dimensions ($JA = 7.54$; $JPK = 7.53$; $JMS = 7.50$). This profile corresponds to students who consistently report implementing critical evaluation strategies. Notably, reasoning style (CRT_{score}) did not

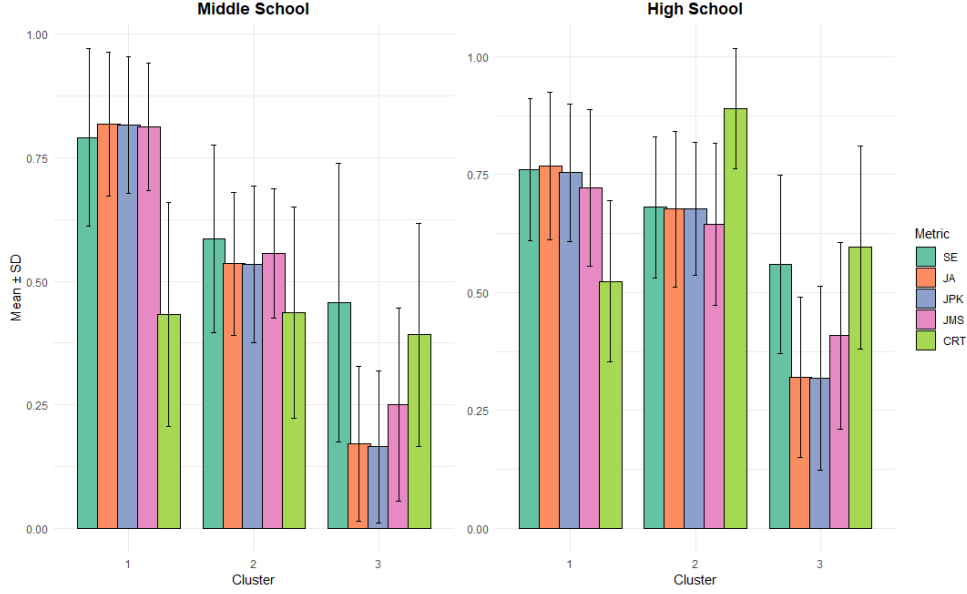


Fig. 2 CTD profiles (cluster) for middle and high school students

Table 5 Means and sample sizes for the different metrics by cluster and school type (SD in round brackets)

School	Cluster	Size	SE	JA	JPK	JMS	CRT _{score}
Middle	1 (High CTD)	281	7.33 (1.44)	7.54 (1.16)	7.53 (1.10)	7.50 (1.03)	3.46 (1.81)
	2 (Medium CTD)	325	5.69 (1.52)	5.28 (1.16)	5.27 (1.28)	5.46 (1.06)	3.49 (1.71)
	3 (Low CTD)	170	4.66 (2.25)	2.38 (1.25)	2.36 (1.23)	3.00 (1.56)	3.14 (1.80)
High	1 (High CTD - System 1)	137	7.11 (1.17)	7.45 (1.23)	7.03 (1.15)	6.77 (1.32)	4.20 (1.38)
	2 (High CTD - System 2)	125	6.41 (1.21)	6.41 (1.31)	6.43 (1.12)	6.16 (1.38)	7.11 (1.03)
	3 (Low CTD)	81	5.48 (1.52)	3.57 (1.36)	3.55 (1.56)	4.26 (1.57)	4.77 (1.73)

significantly differ from the other clusters ($\varepsilon^2 = 0.00$), indicating that analytic versus intuitive reasoning does not discriminate among middle schoolers.

- Cluster 2 (Medium CTD, $n = 325$): shows intermediate levels of SE (5.69) and epistemic justifications (JA = 5.28; JPK = 5.27; JMS = 5.46), reflecting occasional or context-dependent critical practices. This profile may reflect a transitional epistemic stance, in which students have begun to develop awareness of critical evaluation strategies but have not yet consolidated them into stable, habitual practices.
- Cluster 3 (Low CTD, $n = 170$): displays low SE (4.66) and markedly low epistemic justification scores (all < 3.28), indicating a pronounced disengagement from critical evaluation, with the lowest engagement profile relative to Clusters 1 and 2.

Table 6 Kruskal–Wallis test results on the clustering variables, by school type and group. ϵ^2 = effect size; Post-hoc = direction of the pairwise comparisons for significant tests (see the supplementary material in Sect. A for mean and median calculation details).

School	Metric	χ^2	df	p	ϵ^2	Post_hoc
Middle School	SE	230.86	2	< .0001	.30	c1 > c2 > c3
	JA	560.12	2	< .0001	.73	c1 > c2 > c3
	JPk	557.01	2	< .0001	.72	c1 > c2 > c3
	JMS	518.04	2	< .0001	.67	c1 > c2 > c3
	CRT _{score}	5.18	2	ns.	0.00	-
High School	SE	63.32	2	< .0001	.19	c1 > c2 > c3
	JA	167.69	2	< .0001	.49	c1 > c2 > c3
	JPk	162.86	2	< .0001	.48	c1 > c2 > c3
	JMS	103.59	2	< .0001	.30	c1 > c2 > c3
	CRT _{score}	179.12	2	< .0001	.52	c2 > c3 > c1,

4.2.2 Critical thinking disposition profiles among high-school students

Developmental heterogeneity emerges among high school students, with two distinct subtypes of high CTD:

- Cluster 1 (High CTD – System 1, $n = 137$): characterised by high SE (7.11) and elevated epistemic justifications (JA = 7.45; JPK = 7.03; JMS = 6.77), but the lowest CRT_{score} (4.20), indicating a predominant reliance on intuitive and heuristic processing.
- Cluster 2 (High CTD – System 2, $n = 125$): shows slightly lower SE (6.41) and epistemic justifications (JA = 6.41; JPK = 6.43; JMS = 6.16), but the highest CRT_{score} (7.11), reflecting sustained engagement in analytical thinking.
- Cluster 3 (Low CTD, $n = 81$): displays low SE (5.48), low epistemic justifications (all < 4.26), and an intermediate CRT_{score} (4.77), suggesting the absence of a coherent evaluative strategy.

This distinction is absent among middle school students, for whom reasoning style (CRT_{score}) does not discriminate between profiles. This pattern appears to reflect a late-adolescent cognitive specialisation, during which critical thinking becomes contingent upon the interaction between self-efficacy and reasoning style. This developmental shift aligns with a well-documented trajectory: metacognitive capacities and the ability to inhibit heuristic responses (System 1) gradually strengthen throughout adolescence, driven by the maturation of the prefrontal cortex (Lee, Bull, & Ho, 2013; Steinberg, 2005). Unlike younger students, high schoolers increasingly develop the capacity to couple self-efficacy with deliberate analytical processing, a crucial condition for effective critical thinking (Kendeou, Butterfuss, Van Boekel, & O'Brien, 2017). Notably, this specific interaction is absent in middle schoolers, a finding that converges with Hémon et al. (2025). In their variable-centred analysis, they found no significant effect of reasoning style, precisely because their sample consisted of younger adolescents who had not yet reached the developmental stage where analytical reasoning modulates dispositional effects. This indicates that, before late adolescence, critical performance

is predominantly influenced by the total intensity of inclinations, without the moderating role of analytical thinking, a cognitive capacity that only becomes functional with age and neural development.

4.2.3 Relationship between self-efficacy and epistemic justifications

At both educational levels, SE is positively correlated with epistemic justifications (Figure 3, Table 7), indicating that students who feel more confident in their ability to evaluate videos report more frequent engagement in critical evaluation strategies. However, the form of this relationship diverges developmentally:

- In middle school, the relationship is linear: higher SE systematically corresponds to higher use of all epistemic justification strategies, following a clear trend: low SE < medium SE < high SE ($0 < 1 < 2$).
- In high school, the relationship becomes non-linear and more nuanced. For JA and JMS, only students with high SE (level 2) show a significant increase in the use of these strategies, whereas students with medium SE do not differ significantly from those with low SE. In contrast, JPK follows a linear pattern similar to that observed in middle school, suggesting that JPK remains sensitive to SE across developmental levels, independently of the threshold effect observed for multiple-source strategies (Table 8).

This finding suggests that, by late adolescence, self-efficacy alone is no longer sufficient for the mobilisation of more demanding epistemic strategies such as corroboration or deference to authority. A critical threshold of self-efficacy appears necessary, indicating a qualitative shift in how motivation translates into critical practice.

Table 7 Correlation (Kendall's tau) between self-efficacy (recoded) and epistemic justifications.

School	Metric	JA	JPK	JMS
Middle School	disc. SE	0.332***	0.333***	0.407***
High School	disc. SE	0.320***	0.268***	0.294***

* $p < .05$, ** $p < .01$, *** $p < .001$.

4.3 RQ3 - Impact of critical thinking disposition profiles on judgements and decisions

4.3.1 Judgements of epistemic trustworthiness and information credibility

Middle school students exhibit strong inter-cluster differentiation. For all four videos, Cluster 1 (High CTD) rates both sources and information significantly more favourably than Clusters 2 and 3 (Table 9), following a consistent pattern: $c1 > c2 > c3$. In

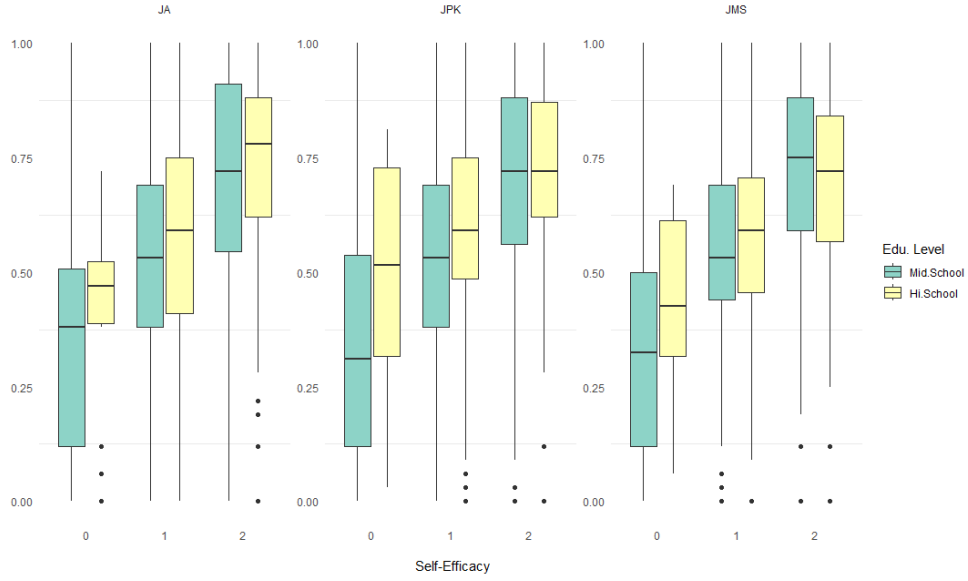


Fig. 3 Distribution of epistemic justifications (JA, JPK, and JMS) by school type and level of self-efficacy. Self-efficacy (SE; 1–9 Likert) was recoded into a three-level ordinal variable, disc.SE: low (1–3), medium (4–6), and high (7–9), coded 0, 1, and 2, respectively.

Table 8 Kruskal–Wallis test results testing the effect of recoded SE on students’ epistemic justifications, by school type. ϵ^2 = effect size; Post-hoc = direction of the pairwise comparisons for significant tests; three self-efficacy groups of disc. SE (0 = low, 1 = medium, 2 = high).

School	Metric	χ^2	df	p	ϵ^2	Post_hoc
Middle School	JA	126.35	2	< .0001	.16	0 < 1 < 2
	JPK	187.80	2	< .0001	.15	0 < 1 < 2
	JMS	120.00	2	< .0001	.24	0 < 1 < 2
High School	JA	52.19	2	< .0001	.15	0 = 1, 2 > 0, 2 > 1
	JPK	36.79	2	< .0001	.11	0 < 1 < 2
	JMS	43.72	2	< .0001	.13	0 = 1, 2 > 0, 2 > 1

contrast, high school students show globally homogeneous judgements, with no significant differences across clusters in most cases. Only two specific contrasts emerge: for V_3 (non-expert, against meat reduction; Table 9), Cluster 2 rates information credibility more negatively than the other clusters; for V_4 (expert, against meat reduction), Cluster 1 judges source trustworthiness more positively. This stability suggests that, beyond a certain developmental threshold, high schoolers base their evaluations on more stable epistemic criteria, being less swayed by motivational fluctuations or dispositional intensity alone.

Table 9 Kruskal–Wallis test results on the information credibility, epistemic trustworthiness, and video-use decision metrics. ε^2 = effect size; Post-hoc = direction of the pairwise comparisons for significant tests (see the supplementary document for calculation details).

Epistemic trustworthiness (ET)						
	Video	χ^2	df	p	ε^2	Post_hoc
Middle School	V ₁	33.61	2	<.001	.04	c1 > c2 > c3
	V ₂	48.40	2	<.001	.06	c1 > c2 > c3
	V ₃	24.01	2	<.001	.03	c1 > c2, c1 > c3, c2=c3
	V ₄	42.00	2	<.001	.05	c1 > c2 > c3
High School	V ₁	0.72	2	ns.	-	-
	V ₂	4.82	2	ns.	-	-
	V ₃	7.20	2	.03	.02	c1 > c2, c2=c3, c1=c3
	V ₄	7.48	2	.02	.02	c1 > c3, c1=c2, c2=c3
Information credibility (IC)						
	Video	χ^2	df	p	ε^2	Post_hoc
Middle School	V ₁	39.46	2	<.001	.05	c1 > c2 > c3
	V ₂	56.59	2	<.001	.07	c1 > c2 > c3
	V ₃	17.40	2	<.001	.02	c1 > c2, c1 > c3, c2=c3
	V ₄	55.19	2	<.001	.07	c1 > c2 > c3
High School	V ₁	0.99	2	ns.	-	-
	V ₂	3.97	2	ns.	-	-
	V ₃	9.16	2	.01	.03	c1 > c2, c3 > c2, c1=c3
	V ₄	6.66	2	.04	.02	ns.
Decision to use the video (SU)						
	Video	χ^2	df	p	ε^2	Post_hoc
Middle School	V ₁	16.33	2	<.001	.02	c1 > c2, c1 > c3
	V ₂	41.78	2	<.001	.05	c1 > c2 > c3
	V ₃	3.15	2	ns.	-	-
	V ₄	35.32	2	<.001	.05	c1 > c2 > c3
High School	V ₁	3.23	2	ns.	-	-
	V ₂	3.69	2	ns.	-	-
	V ₃	3.76	2	ns.	-	-
	V ₄	1.79	2	ns.	-	-

4.3.2 Quality of educational usage decisions

An analysis of decision reliability, operationalised as the tendency to select expert-authored videos, reveals a developmental reversal (Table 10). Among middle school students, Cluster 1 (High CTD) is the most accurate and outperforms Clusters 2 (Medium CTD) and 3 (Low CTD), which do not differ ($c1 > c2 = c3$). Among high school students, Cluster 2 (High CTD – System 2) is the most accurate and outperforms Clusters 1 and 3 ($c2 > c1 = c3$). This pattern suggests that, at the high school level, decision quality depends less on the overall intensity of dispositions than on reasoning style. Notably, high school students in Cluster 1 do not outperform those with low CTD (Cluster 3), illustrating the risks of miscalibrated self-efficacy, a well-documented cognitive trap (Anttonen et al., 2024; Dunlosky & Rawson, 2012) in which

Table 10 Video-use decisions and video sourcing processes among middle school and high school students

Quality of decisions					
School	χ^2	df	p	ε^2	Post-hoc
Middle School	14.08	2	<.001	.02	c1 > c2, c1 > c3, c2 = c3
High School	12.18	2	<.01	.04	c2 > c3, c2 > c1, c1 = c3
Quality of sourcing					
School	χ^2	df	p	ε^2	Post-hoc
Middle School	4.11	2	<i>ns.</i>	—	—
High School	7.49	2	.00	.02	<i>ns.</i>

ε^2 = effect size. Post-hoc = direction of the pairwise comparisons for significant tests.

confidence becomes uncoupled from analytical vigilance. Most importantly, analysis showed a substantial developmental gain. To examine grade-level differences, we conducted two-tailed Wilcoxon rank-sum tests, which show that high school Cluster 2 is significantly more accurate than Cluster 1, the best-performing middle school cluster, with a medium-sized effect ($W = 20,679$, $p < .01$, $r = .22$). This represents the strongest effect observed in our analysis, indicating that the emergence of System 2 reasoning in late adolescence yields markedly better decision quality, despite comparable self-efficacy levels. This underscores that reasoning style, not motivation alone, drives mature critical performance.

4.3.3 Resistance to motivational biases: The role of prior attitude

Beyond judgements based on source expertise, the quality of critical thinking also manifests in the ability to override preexisting beliefs when evaluating counter-attitudinal information. This resistance to confirmation bias, defined as the tendency to favour belief-congruent information, is considered an advanced epistemic competence (Kuhn et al., 2000). We therefore examined whether students' prior stance on meat consumption moderates their evaluations, and whether this sensitivity varies by CTD profile. As shown in Table 11, a nuanced pattern emerges. Among middle schoolers, prior attitude showed a statistically significant, albeit modest, association with judgements of epistemic trustworthiness (Wilcoxon: $W = 119,888$, $p = 0.02$, $r = 0.04$), suggesting a slight tendency to favor videos against meat reduction. This inclination appeared somewhat more evident in Clusters 2 and 3 (medium and low CTD), where students tended to rate against-videos as more credible and trustworthy than pro-videos ($p < 0.05$, $r = 0.06$ – 0.11). This asymmetry suggests that, in the absence of robust critical dispositions, middle schoolers' judgements are guided by their prior beliefs, a pattern consistent with the cognitive superficiality hypothesis in video processing (Salmerón, Sampietro, & Delgado, 2020). In stark contrast, no significant effect of prior attitude is observed among high school students, either globally or within any cluster. This indicates that, by late adolescence, students develop epistemic autonomy, that is, the capacity to decouple source evaluation from ideological preferences. This maturity

coincides with the emergence of the High CTD – System 2 profile (high CRT_{score}), which also demonstrates the highest decision accuracy. It is thus plausible that resistance to confirmation bias is tightly linked to a systematic reliance on analytical reasoning, which enables individuals to suspend judgement and consider alternative viewpoints.

Table 11 Analysis of the effects of students’ attitudes toward the videos’ topic (in favour vs. against) on their judgements

By educational level						
School	Metric	W_s	p	ε	Description	
Middle School	ET	119,888	.02	.04	against > in favour	
	IC	125,949	ns.	–	–	
High School	ET	24,950	ns.	–	–	
	IC	24,373	ns.	–	–	

By educational level and cluster						
School	Metric	Clusters	W_s	p	ε	Description
Middle School	ET	1 (High CTD)	17,965.5	ns.	–	–
		2 (Medium CTD)	20,722.5	.04	.06	against > in favour
		3 (Low CTD)	4,604.5	<.01	.11	against > in favour
	IC	1 (High CTD)	18,563	ns.	–	–
		2 (Medium CTD)	21,966	.047	.07	against > in favour
		3 (Low CTD)	4,926	.034	.06	against > in favour
High School	ET	1 (High CTD - System 1)	3,779	ns.	–	–
		2 (High CTD - System 2)	3,379	ns.	–	–
		3 (Low CTD)	1,528.5	ns.	–	–
	IC	1 (High CTD - System 1)	4,393	ns.	–	–
		2 (High CTD - System 2)	2,822.5	ns.	–	–
		3 (Low CTD)	1,357	ns.	–	–

ε = effect size (Wilcoxon signed-rank test, paired samples; description: explanation of the effect)

5 Discussion

5.1 Development of critical thinking dispositions: progressive maturation (RQ1)

Our findings confirm a significant developmental progression in CTD between early and late adolescence: high school students reported higher levels of self-efficacy, greater reliance on justification by authority (JA) and personal knowledge justification (JPK), and, most notably, a markedly more analytical reasoning style (CRT_{score}). This developmental gap aligns with well-established neurocognitive models: prefrontal cortex maturation during adolescence enhances cognitive regulation, the inhibition of heuristic processing (System 1), and engagement in deliberate, reflective reasoning (System 2) (Lee et al., 2013; Steinberg, 2005). Two results, however, need particular attention. First, the negligible effect size ($\varepsilon^2 = 0.003$) for justification by source multiplicity (JMS) suggests that—even at the high school level—the corroboration strategy,

though institutionally promoted, remains poorly internalised. This finding resonates with (Anmarkrud et al., 2019), who emphasise the cognitive cost of cross-source verification, which often exceeds students’ available resources. Our results indicate that this skill necessitates both motivation and automated cognitive routines, the latter likely emerging solely through regular, guided practice. Second, the absence of a relationship between self-efficacy (SE) and CRT_{score} among middle school students indicates that, at this stage, confidence in one’s evaluative abilities has not yet become coupled with a specific mode of reasoning; a coupling that clearly emerges by late adolescence.

5.2 Critical thinking disposition profiles: Toward a developmental typology (RQ2)

The clustering analysis reveals a cognitive specialisation emerging in late adolescence. Among middle school students, profiles are primarily distinguished by the overall intensity of CTDs: low, medium or high. In contrast, among high school students, two distinct subtypes of “high CTD” emerge, differentiated by reasoning style:

- High CTD – System 1: high self-efficacy (SE) and well-developed epistemic beliefs, but intuitive reasoning;
- High CTD – System 2: slightly lower SE, yet markedly analytical reasoning.

This distinction is absent among middle schoolers, for whom the CRT_{score} does not differentiate between clusters. It reflects an increasing integration between motivation and cognition: by late adolescence, critical thinking is no longer a matter of dispositional intensity alone, but of a specific configuration in which self-efficacy must be coupled with deliberate, analytical reasoning to become fully functional. This finding extends the literature on epistemic development (Kuhn, Black, Keselman, & Kaplan, 2000) by demonstrating that cognitive maturation does not merely enhance the strength of dispositions, but reshapes their very structure—shifting from a unidimensional continuum to a multidimensional, functionally integrated system.

5.3 Impact of profiles on judgements and decisions

5.3.1 The overconfidence trap (RQ3)

Our response to RQ3 underscores the central contribution of this study: the coupling of self-efficacy (SE) and reasoning style (CRT_{score}) determines the quality of decisions but not the score of epistemic judgements. Among middle school students, the High CTD profile (Cluster 1) issues more favourable judgements and makes the most accurate usage decisions. At this developmental stage, motivation alone appears sufficient to discriminate between expert and non-expert sources, likely because the task is perceived as school-related and inherently motivating (though this remains a post-hoc interpretation, as no direct measure of task perception was included in the study).

Among high school students, judgements become remarkably homogeneous, yet decisions diverge sharply: only students in Cluster 2 (High CTD – System 2) demonstrate significantly superior decisional performance. In contrast, students in Cluster 1 (High CTD – System 1), despite high confidence and self-reported use of critical strategies, do not outperform those with low CTD. This developmental reversal is

critical: it illustrates the risks of miscalibrated self-efficacy (Anttonen et al., 2024; Dunlosky & Rawson, 2012). High confidence, when disconnected from analytical reasoning, can foster an illusory epistemic vigilance, a phenomenon that Hémon et al. (2025) could not observe, as their younger sample (Mean age = 13.17) had not yet developed this dissociation, and their variable-centred approach failed to capture the $SE \times (CRT_{score})$ interaction. Finally, the absence of prior attitude effects among high schoolers—in stark contrast to middle schoolers—confirms that the ability to overcome confirmation bias (Nickerson, 1998) emerges in late adolescence, likely due to the combined influence of cognitive maturation and curricular scaffolding.

5.4 Resistance to motivational biases: critical thinking as epistemic self-distancing

The analysis of prior attitude, introduced as a complementary investigation, reveals a decisive developmental divide. Middle school students, particularly those with medium or low CTD, systematically rated videos about meat reduction as more credible, revealing a clear confirmation bias (Nickerson, 1998). This pattern aligns with the cognitive superficiality hypothesis in video processing (Salmerón et al., 2020), where emotions and prior beliefs override argument analysis. High school students, by contrast, showed no sensitivity to their initial attitude, regardless of CTD profile. While the smaller high school sample ($n = 343$) may entail a marginal reduction in statistical power, this sample size remains adequate for detecting effects of moderate magnitude under standard assumptions. The absence of a significant effect is therefore unlikely to be solely attributable to limited power. Rather, it may indicate a genuine capacity among high school participants to evaluate information sources independently of their ideological preferences, a pattern consistent with the development of epistemic autonomy and advanced critical thinking skills (Kuhn et al., 2000).

This shift co-occurs with the emergence of the System 2 profile, which also demonstrates the highest decisional accuracy, suggesting that critical maturity extends beyond monitoring the source to monitoring one’s own cognitive process (Tay, Ryan, & Ryan, 2016). Thus, far from being mere noise, prior attitude serves as a diagnostic marker of critical depth: its neutralisation signals the transition from heuristic judgement to authentic epistemic evaluation. This finding complements related work from the same research project (Hémon et al., 2025), where the authors identified attitude effects on video evaluation but did not examine developmental moderators. Our study demonstrates that only high school students with analytical reasoning skills successfully override this bias, thereby reinforcing our central thesis: effective critical thinking arises not from dispositional intensity alone but from a synergistic integration of motivation and cognition.

6 Pedagogical and methodological implications

The findings of this study yield concrete implications at three interconnected levels: curriculum design, teacher education, and the development of digital pedagogical tools.

689 They align with a growing body of research emphasising that critical thinking educa-
690 tion cannot be one-size-fits-all but must be tailored to learners’ cognitive-motivational
691 profiles (Anmarkrud et al., 2019; Kendeou et al., 2017).

692 6.1 Toward profile-based differentiated instruction

693 Our identification of three distinct critical thinking disposition profiles among middle
694 schoolers and three more nuanced profiles among high schoolers, particularly the two
695 subtypes of “high CTD”, calls for a rethinking of differentiation in digital literacy
696 instruction. Rather than issuing generic prompts such as “evaluate the reliability of
697 this video”, teachers could adapt their interventions based on the learner’s diagnosed
698 profile:

- 699 • For students with low CTD (middle or high school): Strengthen self-efficacy through
700 scaffolded tasks (e.g., low-stakes successions of tasks, such as identify the source
701 → assess its expertise → choose between two videos using guided criteria) paired
702 with immediate, criteria-based formative feedback. These activities follow princi-
703 ple of “micro-successes” (Bandura, 1997), which gradually calibrate self-perceived
704 competence and reduce cognitive overload.
- 705 • For students with high CTD – System 1 (intuitive high schoolers): the instructional
706 priority would be to make salient the mismatch between confidence and analytical
707 rigour and to systematically elicit System 2 processing. Cognitive decentration exer-
708 cises (Kuhn et al., 2000), such as asking students to write a refutation of their chosen
709 video or explain why a peer might reject it, can externalise reasoning and reduce
710 overreliance on intuitive acceptance, fostering engagement of System 2 (analytical
711 thinking).
- 712 • For all students: Systematically promote justification through source multiplicity, a
713 skill found to be underdeveloped even at the high school level. This requires moving
714 beyond the analysis of a single video toward multi-video corroboration tasks, as
715 advocated in collaborative fact-checking approaches (McGrew et al., 2018).

716 These recommendations resonate with conceptual-change perspectives that emphasize
717 developing learners’ metaconceptual awareness, epistemological sophistication, and
718 intentional learning mechanisms (Vosniadou, 2013).

719 6.2 Curricular alignment and teacher professional development

720 The observed developmental trajectory, particularly the maturation of analytical rea-
721 soning (CRT_{score}) and the emergence of distinct CTD profiles in late adolescence,
722 highlights the need for a fine-grained articulation between middle and high school
723 curricula in digital literacy. While the French *socle commun* (2016) and the PIX cer-
724 tification (2019) (Sect. 1) provide a useful framework, they remain insufficient if they
725 fail to differentiate expectations by developmental stage. In middle school, instruc-
726 tion should prioritise the identification of surface-level cues (source expertise, speaker
727 intentions) and the construction of self-efficacy. In high school, the focus should shift
728 toward overcoming cognitive biases (confirmation bias, overconfidence) and cultivating

analytical reasoning. Yet, current teacher education often addresses these competencies only marginally (Wineburg & McGrew, 2019). We therefore propose integrating dedicated modules into teacher training programmes on: (i) critical evaluation of educational videos; (ii) design of differentiated tasks; and (iii) interpretation of learner profiles using learning analytics. As Kennedy, Ely, Thomas, et al. (2012) argue, effective teacher preparation must bridge the didactics of digital media and the cognitive psychology of the digital learner, a dual competence essential for fostering critical engagement in today’s information-rich classrooms.

6.3 Design of digital tools and learning analytics

This study validates the relevance of person-centred approaches in Educational Technology. The use of fuzzy C-Means clustering not only aligns with prior research (Ali et al., 2025), but it also demonstrates the algorithm’s capacity to uncover nuanced dispositional patterns in educational contexts, thereby opening new avenues for the development of intelligent adaptive systems:

- Digital platforms could diagnose students’ critical thinking disposition profiles in real time, using either brief questionnaires or behavioural traces (e.g., interaction logs, pause frequency, source-corroboration behaviour).
- They could then deliver personalised feedback: for instance, to a “High CTD – System 1” student: “You selected this video with confidence. Take two minutes to consider whether an alternative viewpoint exists.”
- Conversational agents (pedagogical chatbots) could be integrated to scaffold specific epistemic strategies, such as JA or JMS (Zawacki-Richter, Marín, Bond, & Gouverneur, 2019), through guided dialogue and reflective prompts.

In this regard, our work contributes to a vision of data-informed critical thinking education (Brazauskienė, 2025), in which learner data do not serve surveillance but rather enable a fine-grained, ethical, and individualised pedagogy aligned with students’ cognitive-motivational configurations.

7 Conclusion, limitations, and future directions

This study aimed to examine how integrated configurations of critical thinking dispositions, combining self-efficacy, epistemic beliefs, and reasoning style, influence not only judgements of information credibility and source trustworthiness but, more crucially, the quality of educational usage decisions (i.e., preference for expert-authored videos) among secondary school students. By adopting a person-centred approach on a large and representative sample ($N = 1,119$), we identified developmentally differentiated CTD profiles between middle and high school students, revealing three main contributions: the identification of distinct dispositional profiles through fuzzy clustering, the demonstration that decision quality in late adolescence depends on the coupling of self-efficacy and analytical reasoning rather than dispositional intensity alone, and the finding that resistance to confirmation bias emerges specifically among high school students with analytical reasoning skills. Methodologically, the use of Fuzzy C-Means

769 clustering represents an original contribution to computers in education research, offer-
770 ing a person-centred alternative to variable-centred approaches that fail to capture
771 the integrated nature of critical thinking dispositions.

772 However, despite the robustness of its methodology, this study reveals some
773 limitations:

774 7.1 Methodological limitations

775 First, all indicators of critical thinking dispositions (CTD) rely on self-report measures.
776 As (Anttonen et al., 2024; Dunlosky & Rawson, 2012) emphasised, self-efficacy is par-
777 ticularly prone to miscalibration relative to actual competence: students may report
778 using critical strategies while applying them only superficially. Future studies should
779 therefore triangulate self-reports with objective behavioural traces—such as video
780 viewing time, pause usage, cross-source corroboration behaviours, or eye-tracking mea-
781 sures focused on epistemic cues (e.g., expertise indicators). Second, the assessment
782 context, conducted in classrooms with an explicit mention of “critical thinking eval-
783 uation”, exposes the study to social desirability bias (Paul, Macedo-Rouet, Rouet,
784 & Stadtler, 2017). Students may have responded according to what they believed
785 teachers or researchers expected. More ecologically valid protocols, for instance, ask-
786 ing students to autonomously search for videos on a simulated YouTube-like platform
787 without explicit reference to “critical thinking”, would mitigate this effect. Third, prior
788 knowledge of the topic (“reducing meat consumption to promote organic farming”) was
789 not introduced in the analysis, although the literature clearly demonstrates that
790 this factor strongly moderates source evaluation competence (Bråten, Britt, Strømsø,
791 & Rouet, 2011). Integrating a factual knowledge pretest and analysing its interac-
792 tion with CTD profiles would constitute a methodological advancement of primary
793 importance. In the present study, prior knowledge of the topic was not controlled
794 for, which may have influenced students’ evaluations in ways that cannot be disen-
795 tangled from the effects of critical thinking dispositions. For instance, students with
796 greater knowledge of agronomy or nutrition may have been better equipped to assess
797 the expertise of the sources, independently of their dispositional profiles, potentially
798 inflating or obscuring cluster differences. Finally, the ecological validity of our findings
799 is constrained by the use of laboratory-produced videos featuring homogeneous actors
800 (white males aged ~40). While necessary for experimental control, this standardis-
801 ation contrasts sharply with the sociodemographic diversity of content creators on
802 YouTube (Li, Bailey, Huynh, & Chan, 2020), limiting generalisability. Moreover, the
803 exclusive use of male presenters prevents examination of gender effects on perceived
804 expertise, a documented bias in source evaluation (Hendriks, Kienhues, & Bromme,
805 2015). Future replications using authentic videos, pre-screened for misinformation by
806 expert raters, would significantly enhance external validity.

807 7.2 Conceptual limitations

808 Two conceptual limitations also deserve attention. First, the person-centred approach
809 provides a cross-sectional snapshot of CTD at a single point in time, precluding

insight into their developmental dynamics. Yet, as P. Facione (1990) notes, these dispositions are inherently malleable and responsive to educational experiences. Future work should employ longitudinal designs or pre-post experimental interventions (e.g., cognitive decentration workshops) to model how profiles evolve and assess their plasticity. Second, our clustering model does not incorporate task-specific motivational variables: interest in the topic, perceived academic stakes, or affective responses to the videos. Yet, such factors strongly modulate cognitive engagement and epistemic processing quality (List, Grossnickle, & Alexander, 2016). Although our supplementary analysis revealed the moderating role of prior attitude—especially among middle schoolers—this variable was not integrated a priori into profile construction. A promising direction would be to enrich future profiling models with measures of subjective task value (Eccles & Wigfield, 2020), including perceived utility, intrinsic interest, and perceived costs, to better capture the full complexity of epistemic judgement in authentic educational contexts.

7.3 Future research directions

Gender was not employed as a discriminant indicator in this study; however, it will be implemented in subsequent investigations as both (i) a clustering covariate to test profile invariance across genders, and (ii) a predictor of profile membership via multinomial logistic regression, to examine potential gender-specific patterns in critical thinking disposition profiles and their predictive power for video evaluation outcomes.

In addition to this, four key avenues emerge for future research. First, studies should examine whether CTD profiles predict not only final decisions but also online processing strategies (eye-fixations, pause usage, rewatching), enabling integrated modelling of epistemic cognition in real time. Second, within a design-based research framework (Anderson & Shattuck, 2012), profile-tailored interventions could be developed and evaluated—e.g., cognitive decentration activities specifically targeting “High CTD – System 1” students. Third, this typology should be extended to other media formats (TikTok, podcasts, infographics), accounting for their distinct modal and cognitive constraints (Sundar et al., 2021). Finally, co-design with teachers could yield practical diagnostic tools (e.g., classroom observation rubrics, behavioural indicators) to operationalise these profiles in daily practice and ensure the ecological transfer of research insights.

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Appendix A Supplementary material

Table A1 Descriptives by cluster, High School.

Statistique	cluster	SE	JPk	JMS	JA	CRT
N	1	137.00	137.00	137.00	137.00	137.00
	2	125.00	125.00	125.00	125.00	125.00
	3	81.00	81.00	81.00	81.00	81.00
Missing	1	0.00	0.00	0.00	0.00	0.00
	2	0.00	0.00	0.00	0.00	0.00
	3	0.00	0.00	0.00	0.00	0.00
Mean	1	0.76	0.75	0.72	0.77	0.52
	2	0.68	0.68	0.65	0.68	0.89
	3	0.56	0.32	0.41	0.32	0.60
Median	1	0.75	0.75	0.72	0.78	0.50
	2	0.69	0.69	0.66	0.69	0.88
	3	0.59	0.31	0.44	0.34	0.63
SD	1	0.15	0.15	0.17	0.16	0.17
	2	0.15	0.14	0.17	0.17	0.13
	3	0.19	0.20	0.20	0.17	0.22
Min	1	0.34	0.41	0.28	0.28	0.00
	2	0.19	0.28	0.25	0.25	0.63
	3	0.03	0.00	0.00	0.00	0.13
Max	1	1.00	1.00	1.00	1.00	0.88
	2	1.00	1.00	1.00	1.00	1.00
	3	0.97	0.75	0.88	0.62	1.00

Table A2 Descriptives by cluster, Middle School.

Statistique	cluster	SE	JPk	JMS	JA	CRT
N	1	281.00	281.00	281.00	281.00	281.00
	2	325.00	325.00	325.00	325.00	325.00
	3	170.00	170.00	170.00	170.00	170.00
Missing	1	0.00	0.00	0.00	0.00	0.00
	2	0.00	0.00	0.00	0.00	0.00
	3	0.00	0.00	0.00	0.00	0.00
Mean	1	0.79	0.82	0.81	0.82	0.43
	2	0.59	0.53	0.56	0.54	0.44
	3	0.46	0.17	0.25	0.17	0.39
Median	1	0.81	0.81	0.81	0.81	0.50
	2	0.56	0.53	0.53	0.53	0.50
	3	0.47	0.12	0.25	0.12	0.38
SD	1	0.18	0.14	0.13	0.15	0.23
	2	0.19	0.16	0.13	0.15	0.21
	3	0.28	0.15	0.20	0.16	0.23
Min	1	0.00	0.25	0.44	0.03	0.00
	2	0.00	0.00	0.16	0.03	0.00
	3	0.00	0.00	0.00	0.00	0.00
Max	1	1.00	1.00	1.00	1.00	1.00
	2	1.00	1.00	1.00	0.97	1.00
	3	1.00	0.59	0.75	0.62	1.00